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TECHNOLOGICAL UNIVERSITY DUBLIN

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BLANCHARDSTOWN

Academic Term	2022-23
Year of Study	Year 2
Semester	Semester 2 – May 2023
Date of Examination	Summer
Time of Examination	

Programme Code	Programme Title	Module Code
TU653	Higher Certificate in Science in Computing in information Technology	COMP H2014
TU757	Bachelor of Science in Computing in Information Technology	COMP H2014
TU860	Bachelor of Science (Honours) in Computing	COMP H2014
TU758	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2014
TU863	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2014

Module Title	Operating Systems

Internal Examiner	**Dr. Kevin Farrell**
External Examiner	**Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Question 1 in Section A is COMPULSORY. Candidates should attempt ALL parts of Question 1 in Section A, and ANY <u>two</u> other questions in Section B.**
3.	**Where a question asks you to use diagrams, to obtain full marks, you MUST provide diagrams.**
4.	**Where a question asks you to use code, to obtain full marks, you MUST provide code.**
5.	**This paper is worth 100 marks. Question 1 is worth 40 marks. All other questions are worth 30 marks each.**
6.	**Please make sure you write the question numbers you answer in the boxes on the front of your answer book.**
7.	**This is a two hour examination.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A: COMPULSORY QUESTION

Question 1: Answer *ALL* parts of this question *(4 marks each part)*

Please write the number of this question on the front of your answer book

a) Explain with the aid of a diagram how linked lists are used to form queues of processes in an Operating System.

b) In the Linux Operating System, what is the difference between an *interruptible sleep state* and an *uninterruptible sleep state*?

c) In a *multi-threaded* process why is the time taken, to perform a *context-switch* between **two** individual threads in the same process, shorter than the time taken to perform a context-switch between **two** *single-threaded* processes?

d) In a particular program, the serial fraction of code is **5%** (i.e. **5%** of the code is not parallelisable). According to *Amdahl's Law* (see below), what is the *speed-up* achievable for this program, when run on a multi-core system with **32** cores?

Amdahl's Law:

$$\text{Speed-Up} \leqslant \frac{1}{S + \frac{(1-S)}{N}}$$

where:

- S = Serial Fraction of Code
- N = Number of CPU Cores

e) Describe the **four** simultaneous conditions necessary in order for *deadlock* to occur.

f) What is a *device driver*?

This question is continued on the next page...

g) List **four** resources in a computer that are managed by an Operating System.

h) A file in a Linux Operating System has the following permissions:

```
rwx r-x r-x
```

What is the *octal* representation of these permissions?

i) Draw a diagram of the One-to-One model of multithreading and briefly describe it.

j) Briefly describe **two** differences between *mutual exclusion locks* and *semaphores* and their associated functions/operations.

(40 Marks Total)

Please now write the number of this question in the box on the front of your answer book.

SECTION B: ANSWER ANY *TWO* QUESTIONS

Question 2

If you answer this question please write the number on the front of your answer book

a) Describe the **two** means of supporting *threads* in an Operating System?

(4 marks)

b) In multi-core programming, distinguish between *data parallelism* and *task parallelism.*

(4 marks)

c) Discuss the use of *RAM* in a computer, under the following headings:

i. The purpose of *RAM*.
ii. The relationship between the *CPU* and *RAM*.
iii. The distinction between *instructions* and *data*.

(6 marks)

d) Describe the *dual-mode* of operation of processes in an Operating System, under the following headings:

i. What it is.
ii. How it is implemented; i.e. how does the transition from one mode to the other take place? Use a diagram to aid your answer for this part.
iii. How it allows the Operating System to protect both itself and other system components?

(16 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.

Question 3

If you answer this question please write the number on the front of your answer book

a) Distinguish between a *Program* and a *Process* in an Operating System.

(2 marks)

b) Draw a labelled diagram showing the structure of a *Process* in main memory. Briefly describe the multiple parts of the process in your diagram. Name a function in the *C* programming language that can be used to allocate memory dynamically.

(8 marks)

c) In relation to a *Process Control Block (PCB)*, explain:

i. What is a PCB?
ii. Briefly describe **four** items it contains
iii. What role does a PCB play in a context switch?

(8 marks)

d) In an Operating System, **five** different processes arrive at time 0 in the order: Process #1, #2, #3, #4, #5. The *low level (short-term)* scheduling algorithm then orders the queue as follows:

Process #	Estimated Run Time (s)	Waiting Time (s)
3	1	0
1	2	1
4	3	3
5	50	6
2	60	56

i. In your opinion, what is the scheduling algorithm used here? **Hints**: It is a *non-pre-emptive* algorithm. Give reason(s) for your answer.
ii. Using the same Process information as above, create a similar table for the First-Come First-Served scheduling algorithm.
iii. Based on the *average wait-time* for this algorithm, is it better than First-Come First Served? Give reason(s) for your answer.

(12 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.

Question 4

If you answer this question please write the number on the front of your answer book

a) What is a *race condition*?

(2 marks)

b) List and describe the **three** conditions necessary to solve the *critical-section problem*. Also, briefly list any assumption(s) about the relative speed of processes associated with the **three** conditions.

(7 marks)

c) The code fragment below, shows a general program/thread pseudo-code for solving the *critical-section problem* using *semaphores*. With reference to the newer (non-Dijkstra/non-traditional) definition of *wait()* and *signal()*, explain how this code operates. Your answer should explain the sequence of events that occur when there are **two or more** threads running in the application, attempting to run the critical section.

```
semaphore S; // Initially, S = 1

wait(S);
critical section
signal(S);
remainder section
```

(8 marks)

d) In relation to the *Dining Philosophers' Problem* of process/thread synchronisation:

i. Describe the Dining Philosophers' Problem
ii. Explain why a synchronisation solution is necessary.
iii. Describe, with the aid of pseudo-code, the synchronisation aspects of the solution.
iv. Outline any advantages/disadvantages your solution may have.

(13 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.